

General Examination of Patients using IOT and AI

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Abstract—A general examination of the patient's current state of health is often part of a general examination. A review of their medical history, a physical exam, and different diagnostic tests are often performed as part of this. In order to evaluate several bodily systems, including the pulse rates, respiratory system, cardiovascular system, body temperature, and blood oxygen, the physical exam often includes inspection, palpation, percussion, and auscultation. Depending on what is thought required, the diagnostic tests may include laboratory testing, imaging investigations, and other specialist examinations. A general examination's objectives are to determine any underlying medical issues and gauge the patient's current state of health. With an improvement in technology and miniaturization of sensors, there have been attempts to utilize the new technology in various areas to improve the quality of human life. One main area of research that has seen an adoption of the technology is the healthcare sector. The people in need of healthcare services find it very expensive; this is particularly true in developing countries. As a result, this project is an attempt to solve a healthcare problem currently society is facing. The main objective of the project was to design a remote healthcare system. It has three main parts. The first part being, detection of patient's vitals using sensors, second for sending data to cloud storage and the last part was providing the detected data for remote viewing. Remote viewing of the data enables a doctor or guardian to monitor a patient's health progress away from hospital premises. The Internet of Things (IoT) concepts have been widely used to interconnect the available medical resources and offer smart, reliable, and effective healthcare service to the patients. Health monitoring for active and assisted living is one of the paradigms that can use the IoT advantages to improve the patient's lifestyle. In this project, I have presented an IoT architecture customized for healthcare applications. The aim of the project was to come up with a Health Monitoring System that can be made with locally available sensors with a view to making it affordable if it were to be mass produced. Hence the proposed architecture collects the sensor data through Arduino microcontroller and relays it to the cloud where it is processed and analyzed for remote viewing. Feedback actions based on the analyzed data can be sent back to the doctor or guardian through Email and/or SMS alerts in case of any emergencies.

Index Terms—General Examination, testing, tests, sensors

I. INTRODUCTION

A patient's general health assessment is a crucial part of preventive medicine and healthcare. It helps medical professionals spot any possible health issues by giving them a general understanding of a patient's physical and mental well-being. A comprehensive evaluation of the patient's medical history, a physical examination, and, if required, numerous diagnostic tests are often included of the assessment. The goal of the examination is to determine the patient's general health state, taking into account their vital signs, outward look, and functional capabilities. Healthcare professionals may make well-informed judgements regarding the patient's treatment and management thanks to the information obtained during a general examination. In order to identify medical disorders early, treat them, and promote overall health and wellbeing, a precise and complete general checkup is essential. A patient's general examination is a thorough assessment of their overall health, encompassing both their physical and emotional well-being. Usually, it entails the following actions:

Review of medical history: The doctor will inquire about the patient's medical background, including any diseases, operations, and present symptoms.

Physical examination: This is a thorough assessment of the head and neck, respiratory, cardiovascular, abdominal, and musculoskeletal systems of the body.

Vital sign assessment: The healthcare professional will take the patient's temperature, blood pressure, pulse, and breathing rate.

Diagnostic testing: To learn more about the patient's health state, the healthcare practitioner may, if required, request a variety of diagnostic tests, including laboratory tests, imaging investigations, or specialty examinations.

The general examination's objectives are to find any hidden health issues and determine the patient's current state of health. The healthcare professional may use this information to decide on the patient's diagnosis, course of therapy, and overall management with confidence. For the purpose of fostering

optimum health and well-being, early diagnosis and treatment of medical disorders, and comprehensive general examinations are essential.

II. RELATED WORK

A crucial component of healthcare is the general examination of patients, and there have been several research and advancements in this area such as remote patient monitoring system. The use of technology to remotely monitor the health of patients, including the use of IoT-based systems to gather and analyze health data. Such developments aim to improve the accuracy, efficiency, and accessibility of the general examination process, while also empowering patients to take a more active role in their own health and well-being.

IoT-based health monitoring systems refer to the collection and analysis of patient data using Internet of Things (IoT) technologies in the healthcare industry. The following is a list of relevant research in this area:

Wearable tech: Several wearable gadgets, such as smartwatches and fitness trackers, employ IoT technology to track a patient's health and send the information to a central system.

Healthcare practitioners may remotely monitor the health of patients with chronic diseases, such as diabetes and heart disease, and notify them of any changes or issues using IoT-based remote patient monitoring devices.

Telemedicine: Through the use of IoT-based telemedicine systems, healthcare professionals may interact with patients and provide medical advice without having to make an in-person visit.

Electronic Health Records (EHR): IoT-based EHR systems let medical professionals safely store, retrieve, and exchange patient health data, including test results and medical records.

Clinical Decision Support Systems (CDSS): IoT-based CDSS provide healthcare practitioners relevant insights and suggestions for patient treatment by using real-time health data from wearable technology and other sources.

These technologies have the ability to increase patient participation and provide individuals more control over their own health while also boosting the quality and efficiency of healthcare delivery.

III. IOT BASED HEALTHCARE KIT

A healthcare kit based on the Internet of Things is a set of networked devices that monitor and manage an individual's numerous health metrics. Typically, the kit has a variety of sensors, data processors, and communication modules that allow it to wirelessly gather and transfer data to a central hub or cloud-based system. Together, these gadgets give a full picture of a person's health state. Healthcare experts may evaluate the gathered data to find patterns, uncover abnormalities, and make individualised suggestions to enhance health outcomes. In addition, patients may use the kit to monitor their own improvement over time and make educated lifestyle decisions.

A. Arduino Microcontroller

Arduino Uno is a microcontroller board based on the ATmega328P. It has 14 digital input/output pins (of which 6 can be used as PWM outputs), 6 analog inputs, a 16 MHz quartz crystal, a USB connection, a power jack, an ICSP header and a reset button. It contains everything needed to support the microcontroller; simply connect it to a computer with a USB cable or power it with a AC-to-DC adapter or battery to get started.



Fig. 1. Arduino microcontroller

B. ECG

ECG records the electrical activity generated by heart muscle depolarizations, which propagate in pulsating electrical waves towards the skin. Although the electricity amount is in fact very small, it can be picked up reliably with ECG electrodes attached to the skin. The full ECG setup comprises at least four electrodes which are placed on the chest or at the four extremities according to standard nomenclature (RA = right arm; LA = left arm; RL = right leg; LL = left leg). Of course, variations of this setup exist to allow more flexible and less intrusive recordings, for example, by attaching the electrodes to the forearms and legs. ECG electrodes are typically wet sensors, requiring the use of a conductive gel to increase conductivity between skin and electrodes.



Fig. 3-ECG

Fig. 2. ECG

C. Temperature Sensor

A temperature sensor is a device that monitors and transforms the temperature of an item or environment to an electrical signal that can be read by a microcontroller or other electronic device. There are several kinds of temperature sensors, however the following are among the most common:

Thermocouples: These sensors employ two distinct metals that, when heated, create a voltage. The voltage is proportional to the difference between the temperatures of the two metals.

RTDs (resistance temperature detectors): These sensors use a substance, such as platinum or nickel, whose resistance

varies with its temperature. The resistance variation is measured and transformed into a temperature value.

Similar to RTDs, these sensors employ a substance whose resistance varies as its temperature changes. Nonetheless, thermistors are constructed from ceramic or polymeric materials and are less costly than RTDs.

Infrared sensors: These sensors measure an object's temperature based on the heat it emits. Often used for non-contact temperature measurement are infrared sensors.

Temperature sensors are used in several applications, including HVAC systems, food safety monitoring, and medical equipment. They are essential for monitoring and managing temperature, a crucial parameter in several processes and systems.

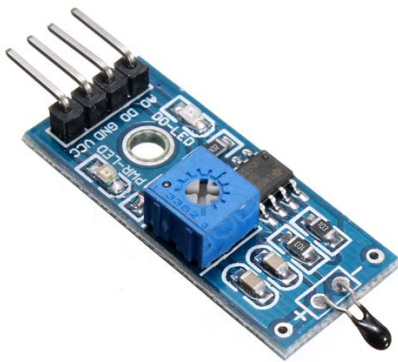


Fig. 3. Temperature sensor

D. Pulse Sensor

A pulse/heartbeat sensor is another essential sensor for IoT devices. Pulse Sensor is an Arduino heart-rate sensor with a well-designed plug-and-play design. Students, artists, sportsmen, makers, and game and smartphone developers can all utilize it to incorporate real heart rate data into their work. With some jumper cables, the sensor attaches to a fingertip or earlobe and connects to the Arduino. It also comes with an open-source monitoring app that displays your pulse graph in real time. Both sides of the sensor are visible. On one side, the LED is paired with a light sensor, while on the other, circuitry is present. On the front, the LED remains. The sensor is also regulated and placed over a vein in our bodies. Regardless of whether it's a fingertip or an ear tip, the top of the vein should be placed. The Pulse Sensor is a plug-and-play heart rate detector for Arduino enabled devices.

E. WIFI Module

The ESP8266 WiFi Module is a self-contained SOC with an inbuilt TCP/IP protocol stack that can provide WiFi access to any microcontroller. The ESP8266 may host an application or outsource all WiFi networking functionality to another processor. Each ESP8266 module comes preprogrammed with an AT command set firmware, so you can just plug it into your Arduino and get about as much WiFi functionality as a WiFi

Shield (right out of the box)! The ESP8266 module is a low-cost board with a large and growing community

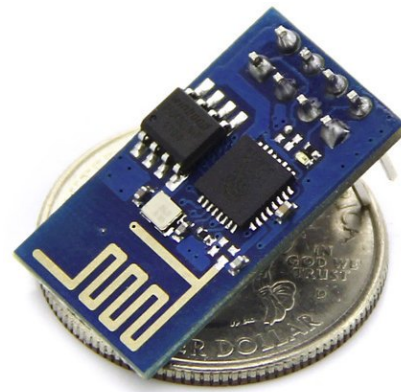


Fig. 4. WIFI module

F. Methodology

Directing IoT-based healthcare and health monitors The framework and sensors will be examined, which monitor the patient's heart rate and core body temperature and then send an email/SMS alert if the values deviate from basic attributes. MySQL database is used to store heartbeat rate and body temperature information so that this system can be controlled from anywhere on the earth over the internet. It also contains a mechanism that may send email/SMS to their family, doctors, and medical specialists in the event of a crisis. Secure data communication is required if we are to develop this connection. This connection's key issue is not only data transmission, but also privacy. A cloud server is also required. This cloud server can be shared by several verified users upon request.

Not only is data transfer a major concern with this link, but so is privacy. You'll also need a cloud server. On request, numerous confirmed users can share this cloud server. Records are updated on a regular basis every 10-15 minutes. In emergency and critical mode, the data is updated every minute. From the wireless gadget, the results will be communicated to the phone using Bluetooth or NFC technology. When you connect to cloud servers, you can acquire results that provide each patient their own address. As a result, before delivering requested information to the patient, received data in the cloud can identify and authenticate the proper patient.

DISCUSSION AND ANALYSIS

We can connect iot-based gadgets to the medical system using the above sensors. We may use this type of system to create a Health Monitoring System that calculates pulse rate and body temperature and monitors the patient's body in real time. This device is capable of sending emergency signals to doctors or patients' relatives. It also helps us save time and money. The review paper's major goal is to connect all types of people with smart healthcare surveillance systems that will provide them with hassle-free medical treatment. Individual

modules utilized in an IoT system, such as pulse detection, temperature detection, and others, yield the following result

EXPECTED RESULTS

With the help of the proposed system the user can monitor and examine the health condition of patients and notify the doctors about results of the patients. By using the system the healthcare professionals can monitor, diagnose, and advice their patients all the time. The health parameters data are stored and published online. Hence, the healthcare professional can monitor their patients from a remote location at any time.

CONCLUSION

By using the system the healthcare professionals can monitor, diagnose, and advice their patients all the time. The health parameters data are stored and published online. Hence, the healthcare professional can monitor their patients from a remote location at any time. Our system is simple. In the designed system the enhancement would be connecting more sensors to internet which measures various other health parameters and would be beneficial for patient monitoring i.e. connecting all the objects to internet for quick and easy access. Establishing a Wi-Fi mesh type network to increase in the communication range. .

REFERENCES

Please number citations consecutively within brackets [1]. The sentence punctuation follows the bracket [2]. Refer simply to the reference number, as in [3]—do not use “Ref. [3]” or “reference [3]” except at the beginning of a sentence: “Reference [3] was the first . . .”

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